MILITARY SPECIFICATION SHEET

POWER CONTROLLER, DC LOAD SWITCHING SPST, NORMAL OPEN, 10 AMPERES

The complete requirements for procuring the controllers described herein shall consist of this document and the latest issue of Specification MIL- .

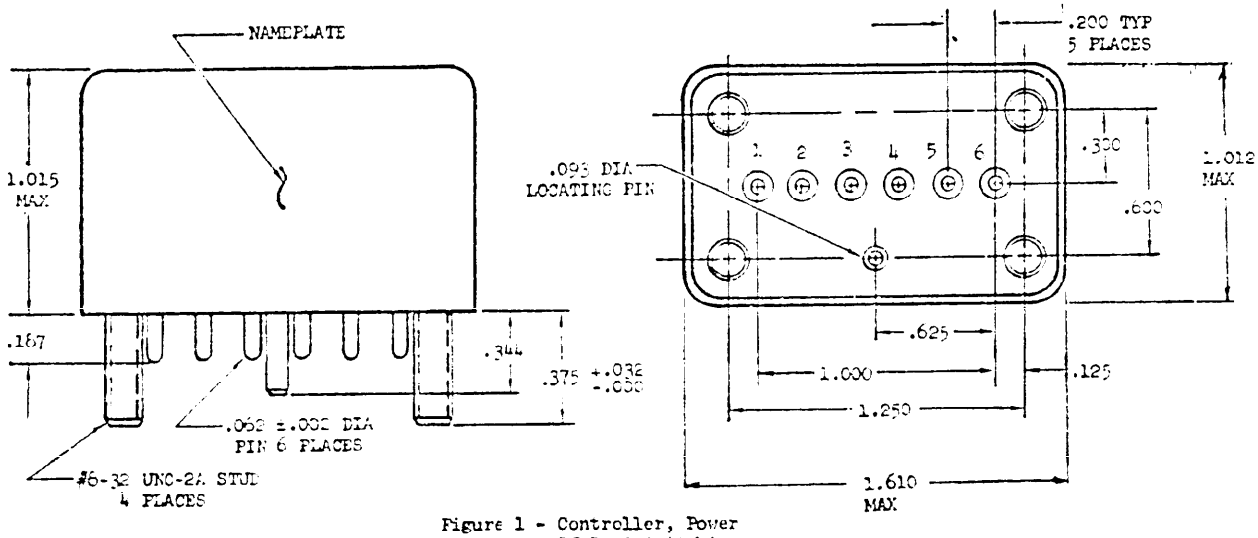


Figure 1 - Controller, Power DC Load Switching

PREFERENCE PIN DIAGRAM
TERMINAL DESIGNATIONS ARE FOR REFERENCE ONLY, DO NOT APPEAR ON HEADER

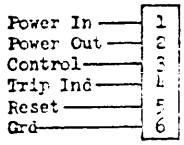


Figure 2 - Terminal Identification

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ± .02 for two place decimals and ± .005 for three place decimals.
3. Terminals shall be tin plated followed by a flow process or coated with composition Sn 40 to Sn 70 solder conforming to QQ-S-571 to facilitate soldering.

REQUIREMENTS:

MECHANICAL AND DIMENSIONAL CHARACTERISTICS

Dimensions and configuration	See Figure
Enclosure	Black anodized metal, sealed-watertight
Terminals	See Figure, Terminals shall be electrically isolated from case.
Weight	3.0 ounces maximum
Thermal resistance - case-to-sink	.50°C/watt max with mounting torque of 15 in-lb

FSC 6110

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ELECTRICAL CHARACTERISTICS (120°C case temperature unless otherwise specified)

Terminal arrangement	SPST (normally open)
Rated operating voltage	29 vdc
Current rating	10.0 amperes
Control voltage	± 10.0 volts dc maximum
Rated control voltage	+ 5.0 volts dc
Turn-on voltage	+ 3.5 volts dc minimum
Turn-off voltage	+ 2.5 volts dc maximum
Turn-on time	1.0 millisecond maximum
Turn-off time	1.0 millisecond maximum
Output rise time	0.1 millisecond minimum 0.5 millisecond maximum
Output fall time	0.1 millisecond minimum 0.5 millisecond maximum
Control input resistance	500 ohms ± 10 percent
Voltage drop	0.5 volts dc maximum @ rated current
Leakage current	1.0 milliampere maximum @ rated voltage
Power dissipation - actuated	8.5 watts maximum
Power dissipation - unactuated	.178 watts maximum
Current limiting	See Figure 1 of MIL-C-XXXXX
Trip-out time	See Figure 2 of MIL-C-XXXXX
Rupture capacity	2000 amperes
Reset voltage	+ 3.5 volts dc minimum + 5.0 volts dc nominal + 10.0 volts dc maximum
Application time to reset	5.0 to 20 milliseconds
Interruption time to reset	5.0 milliseconds minimum
Trip indication voltage	+ 5.0 volts ± 10% @ 10.0 milliamperes
Reset cycle time	30 seconds minimum
Operating voltage transients	MIL-STD 704
Control input transients	100 volts applied between terminals 3 and 6, and 5 and 6
Dielectric withstanding voltage	1,000 volts rms
Fail safe current	30.0 amperes @ 25°C case temperature
Life (operating cycles)	1,000,000 minimum

ENVIRONMENTAL CHARACTERISTICS

Case operating temperatures	-54°C to + 120°C
Case storage temperature	-65°C to + 150°C
Shock	100G for 11 milliseconds
Vibration	30G, 76-2000 Hz
Acceleration	100G
Altitude	Sea level to 100,000 feet

Part number: MXXXXX/9-0001

TABLE I. Dash numbers and applicable characteristics.

Part Number M /9-	Current Rating * (amps)	Power Dissipation (Watts)		Failsafe Current (amps)
		On	Off	
0001	10	8.5	.178	30.0

* Inductive, Capacitive, Resistive or Lamp

Custodian:
Navy - AS

Preparing activity:
Navy - AS
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-P-81653/9(AS) POWER CONTROLLER, DC LOAD SWITCHING SPST, NORMAL OPEN, 10 AMPERES

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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